1. Figures in 'A Simple Geometric Proof of Kepler's First Law' (arXiv:2608.18537)

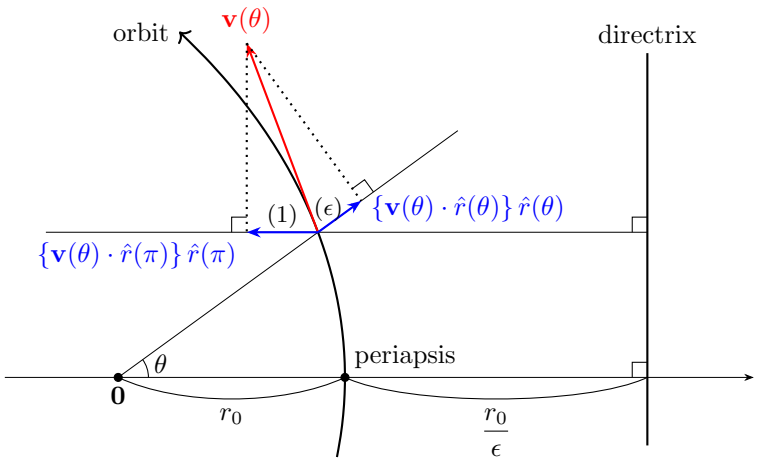


FIGURE 1.1. Kepler's first law: The proof is completely encoded in the fact that the ratio $\dfrac{\mathbf{v}(\theta)\cdot\hat{r}(\theta)}{\mathbf{v}(\theta)\cdot\hat{r}(\pi)} = \epsilon$ is independent of θ.

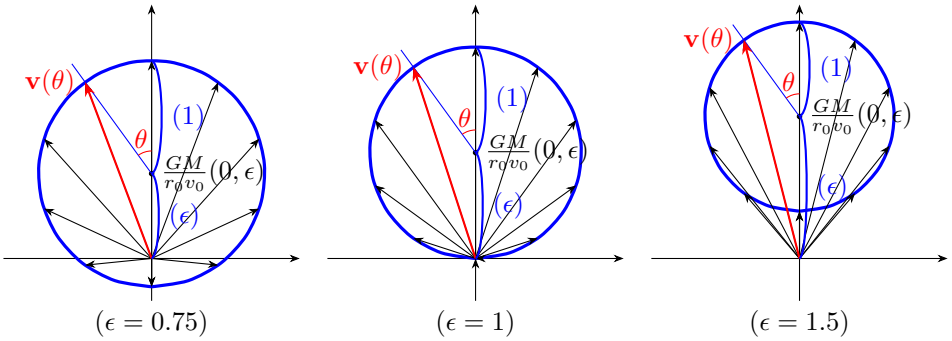


FIGURE 1.2. Hodographs for the cases $\epsilon = 0.75$, $\epsilon = 1$, and $\epsilon = 1.5$.

2. Figures in 'Uniqueness of non-Euclidean Mass Center System and Generalized Pappus' Centroid Theorems in Three Geometries' (arXiv:2412.03080)

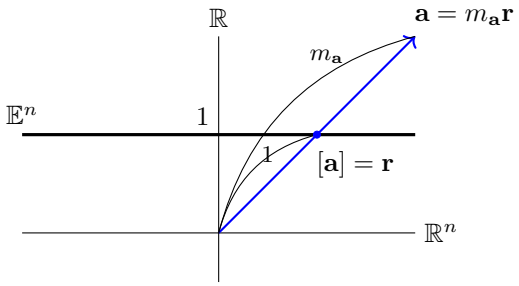


FIGURE 2.1. Euclidean Space $\mathbb{E}^n$ lying on the ambient space $\mathbb{R}^{n+1}$, a material point $[\mathbf{a}]$ with mass $m_{\mathbf{a}}$, and the corresponding material vector $\mathbf{a}$. The concepts of material point, mass, and material vector will be introduced in the following section.

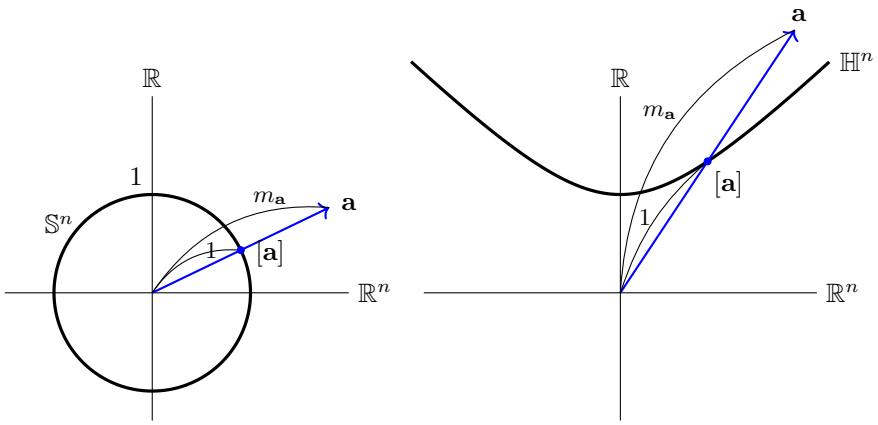


FIGURE 2.2. Spherical Space $\mathbb{S}^n$ and hyperbolic space $\mathbb{H}^n$ lying on the ambient space $\mathbb{R}^{n+1}$, material points $[\mathbf{a}]$ with mass $m_{\mathbf{a}}$, and the corresponding material vectors $\mathbf{a}$

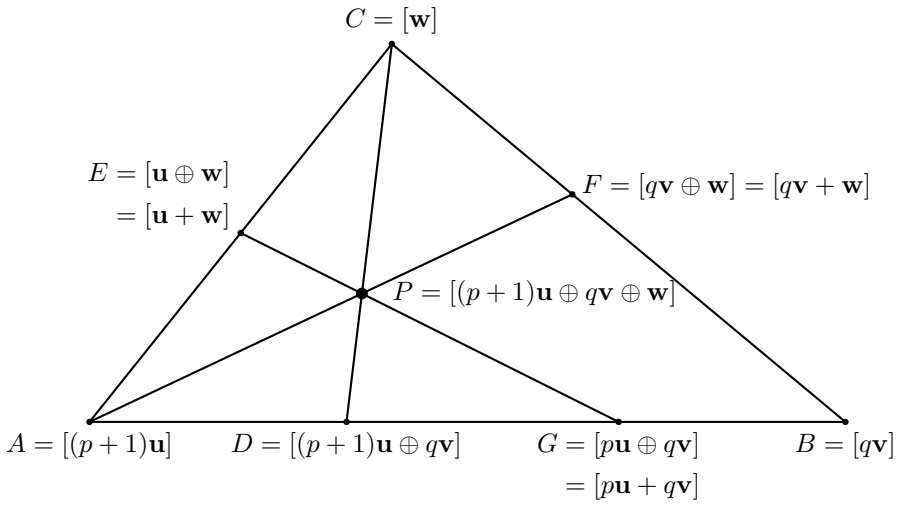


FIGURE 2.3. concurrent segments $\overline{AF}$, $\overline{CD}$, $\overline{EG}$ and the induction hypothesis applied on the points E, F, G

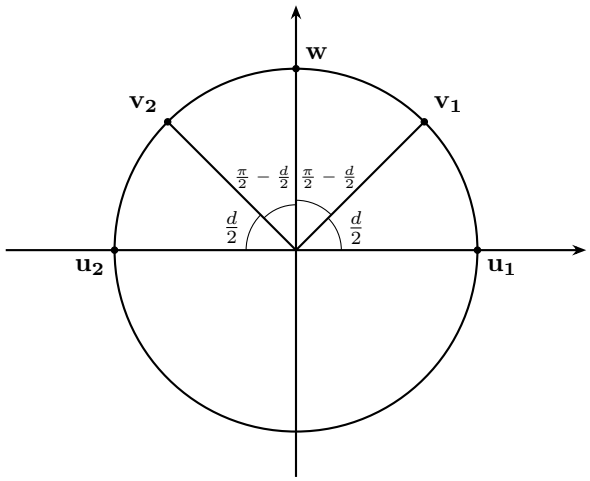


FIGURE 2.4. five material points on $\mathbb{S}^1$

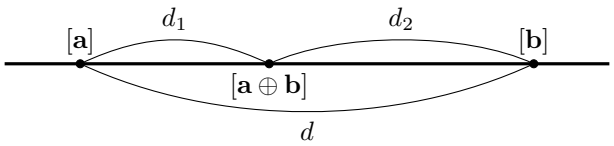
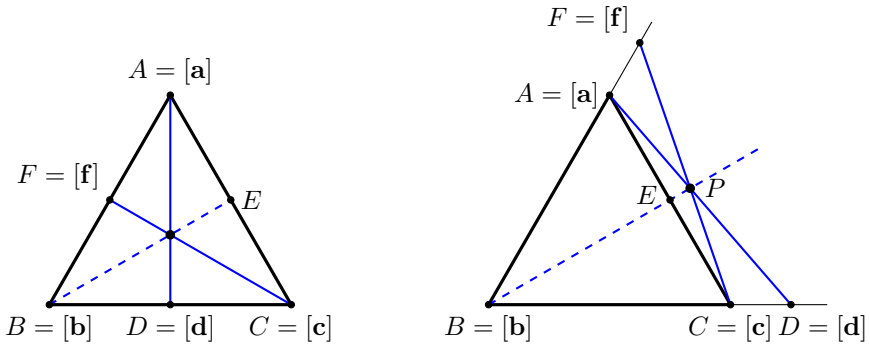
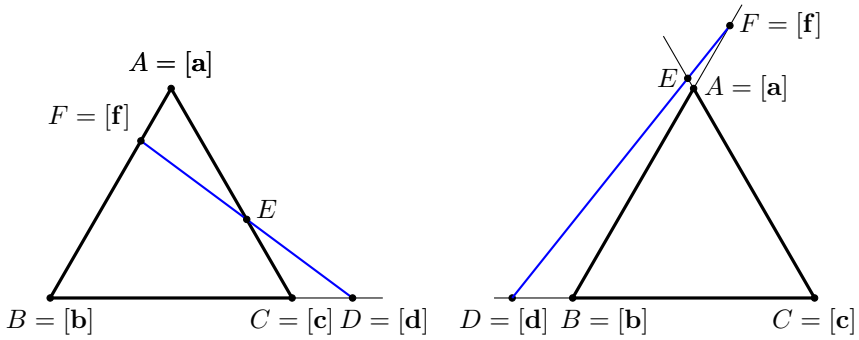


FIGURE 2.5. two material points and their mass center



(A) CASE 1 (B) CASE 2

FIGURE 2.6. [Ceva's theorem] Masses are placed on $A = [\mathbf{a}]$, $B = [\mathbf{b}]$, $C = [\mathbf{c}]$ so that $\mathbf{f} = \mathbf{a} \oplus \mathbf{b}$, $\mathbf{d} = \mathbf{b} \oplus \mathbf{c}$ (CASE 1) or $\mathbf{a} = \mathbf{b} \oplus \mathbf{f}$, $\mathbf{c} = \mathbf{b} \oplus \mathbf{d}$ (CASE 2).



(A) CASE 3 (B) CASE 4

FIGURE 2.7. [Menelaus' theorem] Masses are placed on $A = [\mathbf{a}]$, $B = [\mathbf{b}]$, $C = [\mathbf{c}]$ so that $\mathbf{f} = \mathbf{a} \oplus \mathbf{b}$, $\mathbf{c} = \mathbf{b} \oplus \mathbf{d}$ (CASE 3) or $\mathbf{a} = \mathbf{b} \oplus \mathbf{f}$, $\mathbf{b} = \mathbf{c} \oplus \mathbf{d}$ (CASE 4).

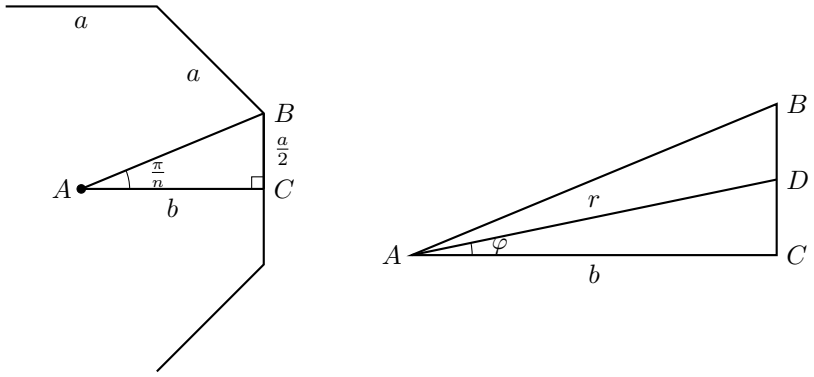


FIGURE 2.8. a spherical or hyperbolic regular n-gon

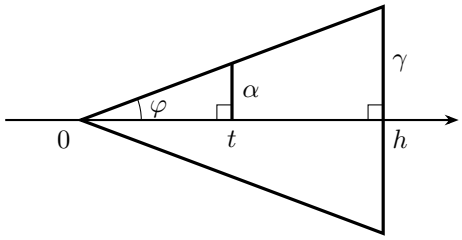


FIGURE 2.9. the triangular cross-section passing through the apex of a right circular cone

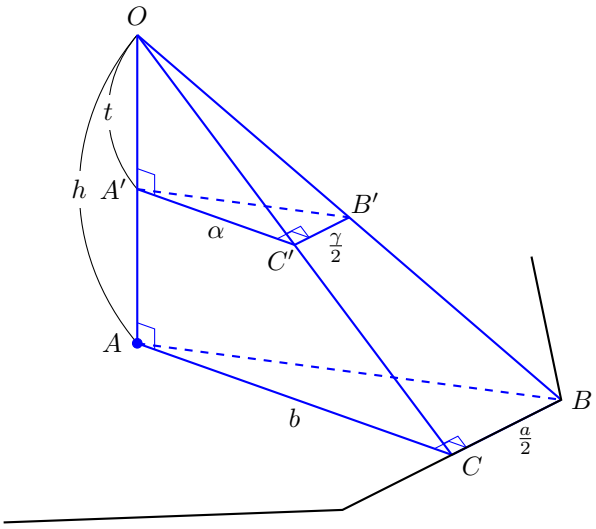


FIGURE 2.10. a spherical or hyperbolic right cone with a regular n-gonal base